

Linear Algebra – Bootcamp

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1 Motivation

Linear algebra can feel a little too abstract in the beginning. Below, we first give some motivation and statistics-related examples before we get into the technical algebraic details.

Premise

Statistics begins with *data*, and data live in *vector spaces*.

- A dataset can be viewed as a *vector*: with n observations, each data point is a point in $\mathbb{R}^n$.
- Statistical methods are often about *transforming* these vectors, for example:

projecting, rotating, decomposing.

- Linear algebra gives us the tools for performing and understanding these transformations.

Example 1.1. Some classical applications of linear algebra in statistics are the following.

1. *Regression is about projection.*

Consider the linear regression model

$$Y = X\theta + \varepsilon, \quad Y \in \mathbb{R}^n, \quad X \in \mathbb{R}^{n \times p}, \quad \theta \in \mathbb{R}^p.$$

The set $\{X\theta : \theta \in \mathbb{R}^p\}$ is a subspace of $\mathbb{R}^n$. The fitted values $\hat{Y}$ are the *projection* of Y onto $\text{col}(X)$, the column space of X , i.e.,

$$\hat{Y} = P_{\text{col}(X)}Y.$$

In other words, regression seeks the closest point in the subspace spanned by the predictors.

2. *PCA finds the best subspace.*

Principal Component Analysis (PCA) takes a high-dimensional data cloud and finds the low-dimensional subspace that captures the maximum variance.

- *Eigenvalues* measure how much variance is explained along each direction.
- *Eigenvectors* define the new axes (principal components) that span this optimal subspace.

Thus, PCA is a dimensionality-reduction technique grounded in linear algebra.

3. Covariance uses inner products.

The covariance between two variables X and Y can be written as

$$\text{Cov}(X, Y) = \frac{1}{n} \langle X - \bar{X}, Y - \bar{Y} \rangle,$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product. Geometrically, covariance encodes the “angle” between two centered data vectors. A large positive covariance means they point in similar directions, while a negative covariance indicates opposite directions.

2 Vector Spaces and Subspaces

Definition 2.1. A *vector space* V over $\mathbb{R}$ is a set equipped with two operations:

1. *Vector addition*: $(u, v) \mapsto u + v \in V$,
2. *Scalar multiplication*: $(r, v) \mapsto rv \in V$ for $r \in \mathbb{R}$,

such that the following axioms hold for all $u, v, w \in V$ and $r, s \in \mathbb{R}$:

$$\begin{aligned} u + v &= v + u && \text{(commutativity)} \\ (u + v) + w &= u + (v + w) && \text{(associativity)} \\ \exists 0 \in V : u + 0 &= u && \text{(additive identity)} \\ \forall u \in V, \exists (-u) \in V : u + (-u) &= 0 && \text{(additive inverse)} \\ r(u + v) &= ru + rv \\ (r + s)u &= ru + su \\ r(su) &= (rs)u \\ 1 \cdot u &= u. \end{aligned}$$

Example 2.1. Plenty of objects commonly used in statistics and probability can be abstracted as vector spaces.

1. $\mathbb{R}^n$ with coordinate-wise addition and scalar multiplication:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}, \quad \lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}$$

2. The set of all real-valued functions on $\mathbb{R}$, with $(f + g)(x) = f(x) + g(x)$ and $(rf)(x) = rf(x)$. These real-valued functions for us will often be *random variables*.

Definition 2.2. A subset $S \subseteq V$ is a *subspace* of V if it is closed under addition and scalar multiplication:

$$\forall s_1, s_2 \in S, s_1 + s_2 \in S, \quad \forall s \in S, r \in \mathbb{R}, rs \in S.$$

Example 2.2. Some common examples of subspaces are:

1. $\{0\} \subseteq V$. Note that 0 here is the 0 of the vector space V and not the 0 of the real numbers.
2. The span of a set $A = \{v_1, \dots, v_n\}$:

$$\text{span}(A) = \left\{ \sum_{i=1}^n r_i v_i : r_i \in \mathbb{R} \right\}.$$

3. The set of all continuous functions on $[-1, 1]$.

We now define three of the most important concepts in linear algebra, that we will often use in statistics theory and applications (often disguised in different terms).

Definition 2.3. 1. A set of vectors $\{u_1, \dots, u_k\}$ is *linearly independent* if

$$c_1 u_1 + \dots + c_k u_k = 0 \implies c_1 = \dots = c_k = 0.$$

Intuitively, each independent vector provides a new direction of information. Dependence, on the other hand, means some redundancy in representation.

2. Given vectors $\{u_1, \dots, u_k\} \subseteq V$, the *span* is defined as

$$\text{span}\{u_1, \dots, u_k\} = \{c_1 u_1 + \dots + c_k u_k \mid c_1, \dots, c_k \in \mathbb{R}\}.$$

The span is the *smallest subspace* of V containing $u_1, \dots, u_k$. For example,

$$\text{span}\{(1, 0), (0, 1)\} = \text{the } xy\text{-plane in } \mathbb{R}^2.$$

3. A set $\{u_i : i \in I\}$ is a **basis** of V if

- (a) the vectors are linearly independent, and
- (b) their span equals V .

Intuitively, a basis is a *minimal set of building blocks*. Every vector in V can be written uniquely as a linear combination of the basis vectors. For example, the standard basis of $\mathbb{R}^2$ is $\{(1, 0), (0, 1)\}$, often denoted as $\{e_1, e_2\}$.

4. The *dimension* of a vector space V , denoted $\dim(V)$, is the number of vectors in any basis of V . Intuitively, the dimension corresponds to the *degrees of freedom* — the number of independent directions available in the space.

5. An *inner product* on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

satisfying for all $u, v, w \in V$ and $a, b \in \mathbb{R}$:

$$\langle u, w \rangle = \langle w, u \rangle \quad (\text{symmetry})$$

$$\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle \quad (\text{linearity})$$

$$\langle u, u \rangle \geq 0, \quad \langle u, u \rangle = 0 \iff u = 0. \quad (\text{positive definiteness})$$

In the special case of two vectors $u, v \in \mathbb{R}^n$, we define

$$\langle u, v \rangle = \sum_{i=1}^n u_i v_i.$$

Intuitively, the inner product measures the *alignment* or similarity between two vectors. Some classical examples of inner products are the following:

- $\langle (1, 0), (0, 1) \rangle = 0$ (orthogonal vectors).
- For functions, $\langle f, g \rangle = \int f(x)g(x) dx$.
- In statistics, covariance can be viewed as an inner product:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)].$$

An extremely important inequality using inner products is the so-called Cauchy–Schwarz inequality:

$$|\langle u, v \rangle|^2 \leq \langle u, u \rangle \langle v, v \rangle.$$

2.1 Problems

- Let A, B be finite subsets of a vector space V such that A is linearly independent and B spans V . Show that

(a) $|A| \leq |B|$,

(b) there exists $B' \subset B$ with $|B'| = |B| - |A|$ such that $A \cup B'$ spans V .¹

Proof. This is widely known as the *Steinitz Exchange Lemma*. We proceed by induction on $m = |A|$. If $m = 0$, the result is trivial.

Suppose the lemma holds for $m - 1$. Let $A = \{u_1, \dots, u_m\}$, $B = \{v_1, \dots, v_n\}$. Since B spans V ,

$$u_m = \sum_{i=1}^{m-1} \alpha_i u_i + \sum_{j=1}^n \beta_j v_j.$$

At least one $\beta_j \neq 0$ (otherwise A would not be independent). WLOG, let $\beta_1 \neq 0$. Then

$$v_1 = \frac{1}{\beta_1} \left(u_m - \sum_{i=1}^{m-1} \alpha_i u_i - \sum_{j=2}^n \beta_j v_j \right),$$

so $v_1 \in \text{span}(A \cup \{v_2, \dots, v_n\})$.

Define

$$C = A \cup \{v_2, \dots, v_n\}.$$

Then $\text{span}(C) = V$. By the inductive hypothesis, $|A| \leq |B|$ and we can find $B' \subseteq B$ with $|B'| = |B| - |A|$ such that $A \cup B'$ spans V . $\square$

- Show that a finite collection of pairwise orthogonal vectors is linearly independent.

Proof. Let $\{u_1, \dots, u_n\}$ be pairwise orthogonal. Suppose

$$c_1 u_1 + \dots + c_n u_n = 0.$$

Taking inner product with u_j gives

$$0 = \langle c_1 u_1 + \dots + c_n u_n, u_j \rangle = c_j \langle u_j, u_j \rangle.$$

Since $\|u_j\|^2 > 0$, it follows that $c_j = 0$ for all j . Thus the set is linearly independent. $\square$

- Let A, B be $m \times n$ matrices. Prove the following properties about the rank of a matrix. To understand inequalities, give examples of specific scenarios where the two sides are equal and others where strict inequality holds.

(a) The row rank of a A is equal to the column rank of a A .

(b) $\text{rank}(A) \leq \min\{m, n\}$.

(c) $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.

(d) $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

(e) $\text{rank}(A) = \text{rank}(A') = \text{rank}(A'A) = \text{rank}(AA')$.

¹Some consequences of this result are that dimension is well-defined and any independent set can be extended to a basis.

Proof. (a) Let $r = \text{col-rank}(A)$. Choose a basis $\{u_1, \dots, u_r\}$ for $\text{col}(A)$ and form $C = [u_1, \dots, u_r] \in \mathbb{R}^{m \times r}$. Each column of A lies in $\text{span}\{u_1, \dots, u_r\}$, so $A = CB$ for some $B \in \mathbb{R}^{r \times n}$. Then $A^\top = B^\top C^\top$, so each column of $A^\top$ is a linear combination of columns of $B^\top$. This shows $\text{row}(A) \subseteq \text{row}(B)$, hence $\dim(\text{row}(A)) \leq r$. Since $\text{rank}(B) = r$, we conclude $\text{row-rank}(A) = r = \text{col-rank}(A)$.

(b) Since $\text{col}(A) \subseteq \mathbb{R}^m$, we have $\text{rank}(A) = \dim(\text{col}(A)) \leq m$. Similarly, we cannot have more independent columns than the total number of columns, so $\text{rank}(A) \leq n$.

Examples: 1. If $A \in \mathbb{R}^{n \times n}$ is square and invertible, then $\text{rank}(A) = n$ (equality). 2. If $A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$, then $\text{rank}(A) = 1 < 2$ (strict inequality).

(c) For any x , $ABx = A(Bx) \in \text{col}(A)$. Thus $\text{col}(AB) \subseteq \text{col}(A)$, and hence $\text{rank}(AB) \leq \text{rank}(A)$. If we denote $B = [b_1, \dots, b_n]$, then $AB = [Ab_1, \dots, Ab_n]$, so $\text{col}(AB) \subseteq \text{col}(B)$. Therefore

$$\text{rank}(AB) = \dim(\text{col}(AB)) \leq \dim(\text{col}(B)) = \text{rank}(B).$$

Hence $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.

Remarks:

- If A is invertible, then $\text{rank}(AB) = \text{rank}(B)$.
- If B is invertible, then $\text{rank}(AB) = \text{rank}(A)$.

Example: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Then $\text{rank}(A) = \text{rank}(B) = 1$, but $\text{rank}(AB) = 0 < 1$ (strict inequality).

(d) For two vector spaces U, V , define their sum $U + V = \{u + v : u \in U, v \in V\}$. Note that $\text{col}(A + B) \subseteq \text{col}(A) + \text{col}(B)$. Taking dimensions:

$$\begin{aligned} \text{rank}(A + B) &= \dim(\text{col}(A + B)) \\ &\leq \dim(\text{col}(A) + \text{col}(B)) \\ &= \dim(\text{col}(A)) + \dim(\text{col}(B)) - \dim(\text{col}(A) \cap \text{col}(B)) \\ &\leq \text{rank}(A) + \text{rank}(B). \end{aligned}$$

(e) From part (a), $\text{rank}(A) = \text{rank}(A^\top)$ since row rank equals column rank.

To show that $\text{rank}(A^\top A) = \text{rank}(A)$, note that $\ker(A) \subseteq \ker(A^\top A)$. For the other inclusion, suppose $x \in \ker(A^\top A)$. Then

$$A^\top A x = 0 \implies \|Ax\|^2 = 0 \implies Ax = 0 \implies x \in \ker(A).$$

Thus, $\ker(A) = \ker(A^\top A)$. By the rank-nullity theorem,

$$\text{rank}(A) = n - \dim(\ker(A)) = n - \dim(\ker(A^\top A)) = \text{rank}(A^\top A).$$

Similarly, $\text{rank}(AA^\top) = \text{rank}(A)$. Therefore,

$$\text{rank}(A) = \text{rank}(A^\top) = \text{rank}(A^\top A) = \text{rank}(AA^\top).$$

□

Intuition

The rank tells us how much *independent information* a matrix carries. In statistics, this is used in topics such as identifiability, model fitting, partitioning of variance, etc.

Example 2.3. (a) *Regression and identifiability.* Consider the model

$$y = X\beta + \varepsilon.$$

Can we estimate β ? This boils down to asking: is X full rank?

- If yes, then $\hat{\beta} = (X^\top X)^{-1} X^\top y$ is well defined.
- If no, then some coefficients are not identifiable.

Property (a) (row rank = column rank) tells us:

Having independent predictors (columns) $\iff$ Having independent equations (rows).

Property (b) ($\text{rank} \leq \min\{m, n\}$) tells us:

We cannot estimate more coefficients than we have data points.

(b) *ANOVA.* The idea is to split variance into meaningful pieces.

Property (c) ($\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$) tells us: combining two stages of linear transformation cannot create more information than was present at the beginning.

Property (d) ($\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$) tells us: the total degrees of freedom used by two explanatory factors together is at most the sum of their individual degrees of freedom. Equality when factors are orthogonal. This is very useful when we use F -tests.

3 Eigenvalues and Eigenvectors

Statistical motivation

Eigenvalues and eigenvectors are the mathematical objects with which we understand the *shape* of variation in data. They play a central role in dimension reduction, PCA, and understanding covariance structure.

Definition 3.1. Let $A \in \mathbb{R}^{m \times m}$ be a square matrix. A non-zero vector $x \in \mathbb{R}^m$ is an *eigenvector* of A if

$$Ax = \lambda x,$$

for some $\lambda \in \mathbb{R}$ called the *eigenvalue* corresponding to x .

In practice, to find eigenvalues we solve the polynomial equation

$$\det(A - \lambda I) = 0.$$

Geometric intuition

Matrices act as linear transformations: they can rotate, stretch, or skew vectors. When a matrix multiplies an eigenvector, it only stretches it (does not rotate it). The stretching factor is exactly the eigenvalue.

3.1 Problems

1. Show that a real symmetric matrix has real eigenvalues.

Proof. Let $A \in \mathbb{R}^{m \times m}$ be a real symmetric matrix, so $A = A^\top$. Take an eigenvalue λ (potentially complex) with eigenvector $v \in \mathbb{R}^m \setminus \{0\}$. Then,

$$\langle Av, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle = \lambda \|v\|^2$$

and

$$\langle v, Av \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle = \bar{\lambda} \|v\|^2.$$

Since A is Hermitian ($A = A^\top$), we have

$$\langle Av, v \rangle = \langle v, Av \rangle.$$

Therefore,

$$\lambda \|v\|^2 = \bar{\lambda} \|v\|^2.$$

Because $v \neq 0$, $\|v\|^2 > 0$, hence $\lambda = \bar{\lambda}$. Therefore, λ is real. □

2. Show that eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal.

Proof. Suppose $A = A^\top$ and $Au = \lambda u$, $Aw = \mu w$ for some $u \neq w$ with $\lambda \neq \mu$. Then

$$\lambda \langle u, w \rangle = \langle \lambda u, w \rangle = \langle Au, w \rangle = \langle u, A^\top w \rangle = \langle u, Aw \rangle = \langle u, \mu w \rangle = \mu \langle u, w \rangle.$$

Therefore,

$$(\lambda - \mu) \langle u, w \rangle = 0.$$

Since $\lambda \neq \mu$, this implies $\langle u, w \rangle = 0$, i.e., u and w are orthogonal. □

3. Show that a square matrix has at most $\text{rank}(A)$ many non-zero eigenvalues.

Proof. Recall that if $A \in \mathbb{R}^{m \times m}$ has $\text{rank}(A) = r$, then there exist matrices $C \in \mathbb{R}^{m \times r}$ and $B \in \mathbb{R}^{r \times m}$ such that $A = CB$, where the columns of C form a basis for $\text{col}(A)$. The key observation² is that the non-zero eigenvalues of A are the same as the eigenvalues of BC . Since BC is an $r \times r$ matrix, it has at most $r = \text{rank}(A)$ non-zero eigenvalues. Therefore, A has at most $\text{rank}(A)$ non-zero eigenvalues. □

4 Spectral Theorem

Let $A \in \mathbb{R}^{n \times n}$ be real and symmetric. We have already shown that the eigenvalues of A are real and that eigenvectors corresponding to distinct eigenvalues are orthogonal. If we consider the diagonal matrix

$$\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

²For a proof of that, suppose $Au = \lambda u$. Then, $Au = \lambda u$, and thus, $CBu = \lambda u$. Set $w = Bu$. Then,

$$BCw = B(CB)u = B(\lambda u) = \lambda(Bu) = \lambda w.$$

Hence, w is an eigenvector of BC with eigenvalue λ .

containing the eigenvalues, and let $Q = [u_1, \dots, u_n]$ be the matrix of corresponding eigenvectors, then

$$AQ = [Au_1, \dots, Au_n] = [\lambda_1 u_1, \dots, \lambda_n u_n] = Q\Lambda,$$

by definition of eigenvalues and eigenvectors. Note that $Q^\top Q = QQ^\top = I_n$, i.e., Q is orthogonal, and hence $Q^{-1} = Q^\top$. Therefore, we have shown that

$$A = Q\Lambda Q^\top.$$

This representation is well-known as the *Spectral Theorem*.

Geometric takeaway

Symmetric matrices do not twist space; instead, they have perpendicular axes (the eigenvectors). Along these axes, they stretch or shrink vectors by real amounts (the eigenvalues).

Example 4.1. Consider the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

To find its eigenvalues, we solve $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

So the eigenvalues are

$$\lambda_1 = 3, \quad \lambda_2 = 1.$$

Now, we find the corresponding eigenvectors. For $\lambda_1 = 3$, we solve

$$(A - 3I)x = 0 \Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} x = 0 \Rightarrow x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Normalize to get $u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. For $\lambda_2 = 1$, we solve

$$(A - I)x = 0 \Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} x = 0 \Rightarrow x_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Normalize to get $u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. We form the matrices

$$Q = (u_1 \ u_2) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

The Spectral Theorem guarantees that

$$A = Q\Lambda Q^\top,$$

which we can also verify manually. □

5 Singular Value Decomposition

The Singular Value Decomposition (SVD) is one of the most important results in linear algebra. It generalizes eigendecomposition and provides a way to “x-ray” a matrix, breaking it into simpler pieces that reveal its geometry.

Theorem 5.1. *Let $A \in \mathbb{R}^{n \times p}$. Then there exist an orthogonal matrix $U \in \mathbb{R}^{n \times n}$, an orthogonal matrix $V \in \mathbb{R}^{p \times p}$, and a diagonal rectangular matrix $\Sigma \in \mathbb{R}^{n \times p}$ with nonnegative entries $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ on the diagonal (the singular values), such that*

$$A = U\Sigma V^\top.$$

Geometric intuition

Geometrically, orthogonal matrices U and V represent rotations (or reflections), while the diagonal matrix Σ represents scaling along coordinate axes. SVD thus expresses a matrix as a composition of three transformations: rotate $\rightarrow$ scale $\rightarrow$ rotate. This allows us to understand how a matrix stretches and rotates vectors.

SVD is used extensively in applications such as image compression, signal processing, low-rank approximation, etc.

Example 5.1. We derive the SVD of the matrix

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

First, we compute $X^\top X$ and its eigenpairs:

$$X^\top X = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Solve $\det(X^\top X - \lambda I) = 0$:

$$\det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) = 0.$$

Hence, the eigenvalues are $\lambda_1 = 3$, $\lambda_2 = 1$. Next, we find singular values and right singular vectors V . The singular values are $\sigma_i = \sqrt{\lambda_i}$:

$$\sigma_1 = \sqrt{3}, \quad \sigma_2 = 1.$$

Eigenvectors of $X^\top X$:

$$\lambda = 3: (X^\top X - 3I)v = 0 \Rightarrow v \propto \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$\lambda = 1: (X^\top X - I)v = 0 \Rightarrow v \propto \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Thus,

$$V = (v_1 \ v_2) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 2}.$$

The left singular vectors U are computed via $u_i = \frac{1}{\sigma_i} X v_i$. To see why this is true, recall that $X = U \Sigma V^\top$. Since v_i is an eigenvector, we have

$$X v_i = U \Sigma V^\top v_i = U \Sigma e_i = \sigma_i (U e_i) = \sigma_i u_i,$$

since the orthonormality of the eigenvectors yields

$$V^\top v_i = \begin{pmatrix} v_1^\top \\ \vdots \\ v_p^\top \end{pmatrix} v_i = \begin{pmatrix} v_1^\top v_i \\ \vdots \\ v_p^\top v_i \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = e_i.$$

Thus, we have to compute $X v_1$ and $X v_2$:

$$X v_1 = X \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad u_1 = \frac{1}{\sqrt{3}} X v_1 = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \sqrt{\frac{2}{3}} \end{pmatrix}.$$

$$X v_2 = X \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad u_2 = \frac{1}{\sigma_2} X v_2 = X v_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}.$$

Since X is 3×2 , $U \in \mathbb{R}^{3 \times 3}$ must be orthogonal. We choose u_3 orthonormal to u_1, u_2 . One convenient choice is

$$u_3 = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{pmatrix}.$$

Then $U = (u_1 \ u_2 \ u_3)$ is orthogonal. Assembling everything, SVD gives $X = U \Sigma V^\top$, with

$$U = \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \sqrt{\frac{2}{3}} & 0 & -\frac{1}{\sqrt{3}} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

□

Checks and interpretation.

- $\text{rank}(X) = 2$ (two positive singular values).
- Columns of V (right singular vectors) give orthonormal directions in input space $\mathbb{R}^2$.
- Columns u_1, u_2 (left singular vectors) are the corresponding output directions; u_3 spans $\ker X^\top$.
- Energy: $\|X\|_F^2 = \sigma_1^2 + \sigma_2^2 = 3 + 1 = 4$. Best rank-1 approximation keeps only $\sigma_1 u_1 v_1^\top$.

6 Examples and Applications

1. Consider the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Find its eigenvalues and orthonormal eigenvectors, and then show that the unit circle

$$\{x \in \mathbb{R}^2 : \|x\|_2 = 1\}$$

is mapped by A to an ellipse aligned with the eigenvectors.³

Solution. Compute eigenvalues by solving $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = (\lambda - 1)(\lambda - 3).$$

Hence, $\lambda_1 = 1$, $\lambda_2 = 3$. We find the eigenvectors:

$$\lambda_1 = 1 : \quad (A - I)x = 0 \quad \Rightarrow \quad x_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

$$\lambda_2 = 3 : \quad (A - 3I)x = 0 \quad \Rightarrow \quad x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Let

$$Q = (q_1 \ q_2) = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Then, by the spectral theorem, $A = Q\Lambda Q^\top$. Now take $x \in \mathbb{R}^2$ with $\|x\|_2 = 1$. Since Q is orthogonal, we can write $x = Qz$ for some z with $\|z\|_2 = 1$. Then

$$y = Ax = Q\Lambda Q^\top x = Q\Lambda z.$$

In the eigenbasis (coordinates given by $Q^\top$), we compute

$$y' := Q^\top y = Q^\top (Q\Lambda z) = \Lambda z.$$

Writing $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$, this becomes

$$y' = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} z_1 \\ 3z_2 \end{pmatrix}.$$

Since $z_1^2 + z_2^2 = 1$, the points y' satisfy

$$\left(\frac{y'_1}{1}\right)^2 + \left(\frac{y'_2}{3}\right)^2 = 1,$$

which is the equation of an ellipse with semi-axes 1 and 3, aligned with the eigenvectors q_1 and q_2 . □

2. Let $X \in \mathbb{R}^{n \times p}$ be a data matrix (rows=observations, columns=variables) with sample covariance

$$S = \frac{1}{n} X^\top X.$$

³This problem illustrates nicely the geometry imposed by the spectral theorem.

- (a) Show that for any unit vector $v \in \mathbb{R}^p$, the variance of the projected data Xv is $v^\top S v$.
 (b) Show that the direction of maximum variance is the eigenvector of S with the largest eigenvalue.

This problem gives us a direct bridge to the famous PCA algorithm that is commonly used in applied statistics.

Proof. (a) For $v \in \mathbb{R}^p$ with $\|v\| = 1$, consider the projected data $Xv \in \mathbb{R}^n$. Its sample variance is

$$\text{Var}(Xv) = \frac{1}{n}(Xv)^\top (Xv) = \frac{1}{n}v^\top X^\top X v = v^\top S v.$$

- (b) Recall that for any eigenpair (λ, v) of S with $\|v\| = 1$, we have

$$v^\top S v = \lambda.$$

From part (a), $v^\top S v$ is the variance in direction v . Maximizing over all unit v gives

$$\max_{\|v\|=1} v^\top S v = \lambda_{\max}(S).$$

Any maximizer v^* satisfies $Sv^* = \lambda_{\max}(S)v^*$. Hence, the first principal component direction is the top eigenvector of S , and the maximum variance equals $\lambda_{\max}(S)$. $\square$

3. Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Show that the following are equivalent:

- (a) A is positive semidefinite, i.e., $x^\top A x \geq 0$ for every $x \in \mathbb{R}^n$.
 (b) All eigenvalues of A are non-negative.

Proof. (i) $\implies$ (ii): Let v be an eigenvector of A with eigenvalue λ . Then

$$v^\top A v = v^\top (\lambda v) = \lambda \|v\|^2.$$

By assumption, $v^\top A v \geq 0$, and since $\|v\|^2 > 0$, it follows that $\lambda \geq 0$.

(ii) $\implies$ (i): By the spectral theorem, $A = Q\Lambda Q^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ and Q orthogonal. For any x , write $x = Qy$, then

$$x^\top A x = y^\top \Lambda y = \sum_{i=1}^n \lambda_i y_i^2 \geq 0,$$

since each $\lambda_i \geq 0$. $\square$

4. Let S be symmetric positive definite. Consider the set

$$E = \{x \in \mathbb{R}^p : x^\top S x \leq 1\}.$$

Show that E is an ellipse aligned with the eigenvectors of S , with radii proportional to $\sqrt{\lambda_i}$.

Proof. Since S is symmetric positive definite, write $S = Q\Lambda Q^\top$ with eigenvalues $\lambda_i > 0$. For $x = Qy$ with $\|y\| = \|x\|$, we have

$$x^\top S x = y^\top \Lambda y = \sum_{i=1}^p \lambda_i y_i^2.$$

Thus the set is

$$E = \left\{ y : \sum_{i=1}^p \lambda_i y_i^2 \leq 1 \right\}.$$

This is an ellipse in the y -coordinates, aligned with the eigenvectors of S . The i -th axis length is proportional to $1/\sqrt{\lambda_i}$. $\square$

5. Let $X \in \mathbb{R}^{n \times p}$ be a full-rank design matrix. Show that the sensitivity of the OLS solution

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

to perturbations in y depends on the minimum eigenvalue of $X^\top X$. Interpret what happens when $\lambda_{\min}(X^\top X)$ is very small. (Hint: multicollinearity).

Proof. Define

$$\hat{\beta}(y) = (X^\top X)^{-1} X^\top y.$$

For a perturbation $y \mapsto y + \Delta y$, the change in the OLS solution is

$$\Delta\beta = \hat{\beta}(y + \Delta y) - \hat{\beta}(y) = (X^\top X)^{-1} X^\top \Delta y.$$

The SVD of X gives $X = U\Sigma V^\top$, where $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_p)$ with $\sigma_1 \geq \dots \geq \sigma_p > 0$. Then,

$$(X^\top X)^{-1} X^\top = V\Sigma^{-1}U^\top,$$

and hence,

$$\Delta\beta = V\Sigma^{-1}U^\top \Delta y.$$

Since U, V are orthogonal, $\|U\|_2 = \|V\|_2 = 1$ and $\|\Sigma^{-1}\|_2 = 1/\sigma_{\min}(X)$. Thus,

$$\|\Delta\beta\|_2 \leq \frac{1}{\sigma_{\min}(X)} \|\Delta y\|_2.$$

Note that $\lambda_{\min}(X^\top X) = \sigma_{\min}^2(X)$, since

$$X^\top X = V\Sigma^\top U^\top U\Sigma V^\top = V(\Sigma^\top \Sigma)V^\top$$

and $X^\top X = \text{diag}(\sigma_1^2, \dots, \sigma_p^2)$. Therefore,

$$\|\Delta\beta\|_2 \leq \frac{1}{\sqrt{\lambda_{\min}(X^\top X)}} \|\Delta y\|_2.$$

Interpretation. If $\lambda_{\min}(X^\top X)$ is small, then $\kappa(X^\top X)$ is large. In this case, small perturbations in y can cause very large changes in $\hat{\beta}$. This happens when the columns of X are nearly linearly dependent, a phenomenon known as *multicollinearity*. In such situations, the OLS estimates become unstable, with large variances. $\square$

6. For a convex quadratic function

$$f(x) = \frac{1}{2}x^\top Ax - b^\top x,$$

with A symmetric positive definite, show that there exists a unique minimizer and derive it.

Proof. Since A is symmetric positive definite, f is strictly convex, so it has a unique minimizer.

To find it, we compute the gradient. Recall:

- For a vector $x \in \mathbb{R}^n$, $\nabla(b^\top x) = b$.
- For a quadratic form, if A is symmetric then

$$\nabla\left(\frac{1}{2}x^\top Ax\right) = Ax.$$

This can be seen by writing $x^\top Ax = \sum_{i,j} a_{ij}x_i x_j$ and differentiating term by term, or by using the matrix calculus identity $\nabla(x^\top Ax) = (A + A^\top)x$. When $A = A^\top$, this simplifies to $2Ax$, and with the prefactor $\frac{1}{2}$ we obtain Ax .

Therefore,

$$\nabla f(x) = Ax - b.$$

Setting the gradient equal to zero gives the first-order condition:

$$Ax = b.$$

Since A is positive definite, it is invertible, so the unique solution is

$$x^* = A^{-1}b.$$

□

7. Show that for any symmetric positive semidefinite matrix S ,

$$\det(S) = \prod_{i=1}^p \lambda_i,$$

where λ_i are its eigenvalues. Interpret this result for a covariance matrix: what does $\det(S)$ measure?

Proof. By the spectral theorem, $S = Q\Lambda Q^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$. Since the determinant is invariant under orthogonal transformations,

$$\det(S) = \det(Q\Lambda Q^\top) = \det(Q) \det(\Lambda) \det(Q)^{-1} = \det(\Lambda) = \prod_{i=1}^p \lambda_i.$$

Interpretation. If S is a covariance matrix, λ_i represent the variances along principal component directions. Thus, $\det(S)$ is the product of the variances, thus tells us how “spread out” the data cloud is in all directions simultaneously. □